EEE 304, HW7

Problem 1.
An AM modulation process uses a carrier frequency of 100kHz. Suppose that the modulated signal is used to modulate a 300kHz carrier. Find the frequencies of the demodulation process that can be used to recover the original signal.

The modulation creates signal replicas centered at 100kHz and -100kHz (assuming that the signal maximum frequency does not create overlaps). The second modulation creates replicas of the original signal at +/-200kHz and +/-400kHz. Hence, a demodulation can use either the 200kHz frequency or the 400kHz frequency.

Problem 2.
Audio signals, bandlimited to 20kHz, are sampled and time-division multiplexed using PAM. What is the minimum frequency of the PAM pulses to enable the simultaneous transmission of 20 audio signals?

Audio signals should be sampled at 40kHz or more, so each signal should be sampled every 25us (or less). In that period we should fit 20 signals and 1 sync signal, so the PAM pulse should be at most 25/21 us = 1.19us, or the minimum PAM frequency is 840kHz.

Problem 3.
Sketch the frequency domain representation of the intermediate signals involved in the "Chopper Amplifier" application. If the chopper frequency is 100kHz, and the chopper duty cycle is 100D% (or, its fraction is D, a number 0-1), determine the signal amplification as a function of the bandpass amplifier A and the duty cycle D.

The sequence of operations in the Chopper Amplifier is

$$x_m(t) = s(\omega_c t)x(t)$$

$$H_{amplify-bandpass}[x_m] = Ax_m(t)$$

$$x_d(t) = s(\omega_c t)Ax_m(t)$$

$$x_a(t) = H_{lowpass}[x_d]$$

The square wave $s(\omega_c t)$ has a Fourier transform

$$S(j\omega) = \sum \frac{2\sin k\omega_c T_1}{k}\delta(\omega - k\omega_c)$$

The duty cycle of the square wave affects the coefficient of the impulses as $T_1 = D\frac{1}{2}T \Rightarrow \omega_c T_1 = \pi D$

$$S(j\omega) = \sum \frac{2\sin k\pi D}{k}\delta(\omega - \omega_c)$$

Then,

$$X_m(j\omega) = \frac{1}{2\pi}X(j\omega) * S(j\omega) = \sum \frac{\sin k\pi D}{k\pi}X(\omega - k\omega_c)$$

We bandpass-amplify

$$X_{bp-a}(j\omega) = H_{BP}(j\omega)X(j\omega) = \sum_{k=\pm 1} A\frac{\sin k\pi D}{k\pi}X(\omega - k\omega_c)$$

Then, modulate again

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$$X_d(j\omega) = \frac{1}{2\pi} X_{bp-a}(j\omega) * S(j\omega) = \sum \frac{\sin k\pi D}{k\pi} X_{bp-a}(\omega - k\omega_c)$$

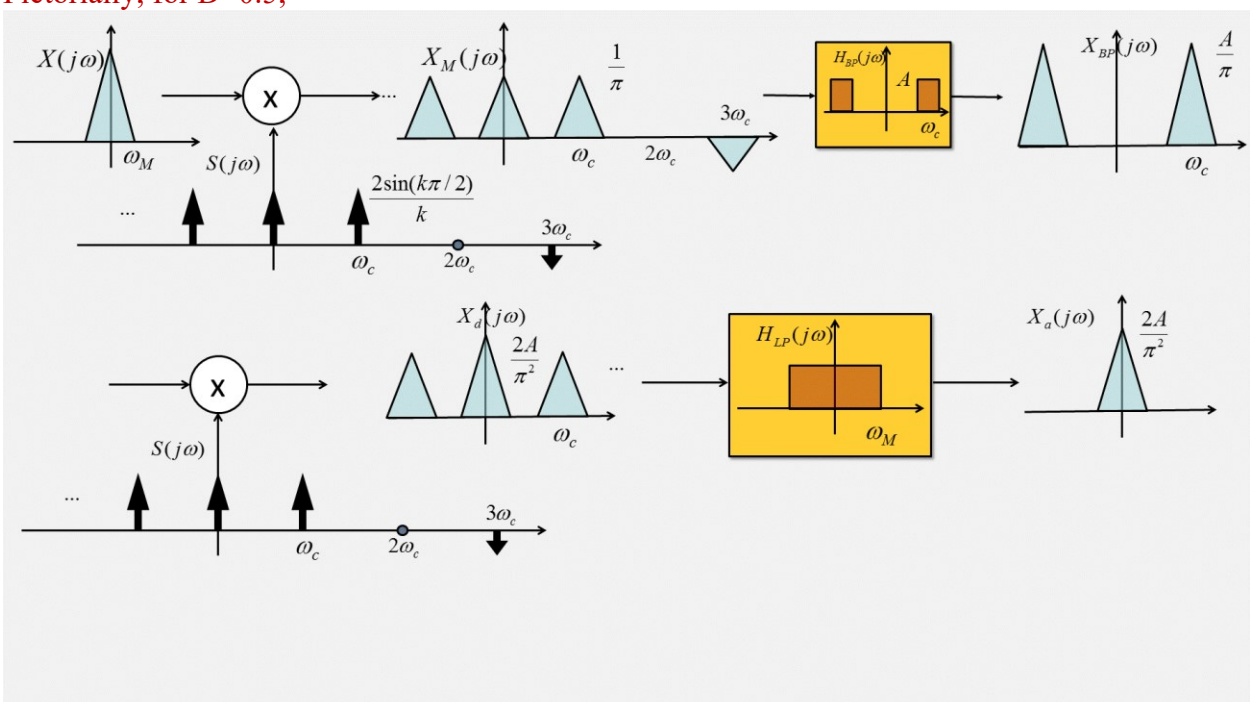
The signal $X_d(j\omega)$ will have two replicas centered at zero frequency, corresponding to k = +1, -1. The rest of the replicas have amplitudes that can be determined but are irrelevant because they will be eliminated after low-pass filtering.

$$X_d(j\omega) = \frac{2A\sin^2 \pi D}{\pi^2} X(j\omega) + \sum_{k\neq 0} a_k X(\omega - k\omega_c)$$

Finally,

$$X_a(j\omega) = H_{LP}(j\omega) X_d(j\omega) = \frac{2A\sin^2 \pi D}{\pi^2} X(j\omega)$$

Pictorially, for D=0.5,



Thus, the requested amplifier gain function is $\frac{2A\sin^2 \pi D}{\pi^2}$.

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